Instrument and control valve maintenance history

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This extract lists the work orders on the instruments and control valves of the basic control loops that have a known performance problem. Each entry gives the work order, the date, what was found and what was done.

FV-1003 Recycle flow valve (loop FIC-1003, U-200)

Work order	Date	Found	Done
WO-24-0417	2024-03-12	Routine positioner check	Positioner calibrated, travel deviation 0.4 %
WO-25-1182	2025-05-20 (turnaround)	Small stem leak at the packing	Graphite packing tightened by two flats
WO-25-1630	2025-08-04	Operators report flow steps of 4 to 6 m^3/h	Positioner diagnostics: friction high, dead band 2.8 % of travel. Packing loosened half a flat, leak reappeared, tightened again. Repacking recommended
WO-26-0211	2026-02-17	Flow cycling with saw-tooth output	Not executed, needs unit at low rate. Planned for next opportunity

Remarks: friction rose after the packing was tightened in WO-25-1182. A low-friction packing (PTFE V-ring) is compatible with the service and is proposed for the next turnaround.

FV-1102 Reflux flow valve (loop FIC-1102, U-100)

Work order	Date	Found	Done
WO-24-0933	2024-07-02	Stem scratched, deposits	Stem polished, packing replaced (graphite)
WO-25-1544	2025-07-15	Reflux flow moves in steps	Travel test: stem moves after 2.4 % output change. Positioner

Work order	Date	Found	Done
			recalibrated, little improvement

FV-1206 Quench flow valve (loop FIC-1206, U-200)

Work order	Date	Found	Done
WO-25-0388	2025-02-26	Slow, stepwise response	Actuator spring checked, supply pressure low (3.1 bar instead of 4.0). Regulator replaced
WO-26-0402	2026-04-08	Steps return	Travel test: dead band 2.2 %. Packing adjustment planned

FV-1214 Side draw flow valve (loop FIC-1214, U-100)

Work order	Date	Found	Done
WO-25-1101	2025-05-14 (turnaround)	Packing replaced	New graphite packing
WO-26-0577	2026-05-19	Side draw cycles with constant setpoint	Travel test: dead band 3.0 %. Friction high since the packing change

TT-1209 Furnace outlet thermocouple (loop TIC-1209, U-100)

Work order	Date	Found	Done
WO-25-0712	2025-04-02	Reading 2.5 °C above the portable reference	Thermocouple replaced (type K)
WO-25-1977	2025-10-09	Reading 1.8 °C above the reference	Calibration interval reduced to 6 months
WO-26-0688	2026-04-15	Reading 2.1 °C above the reference	Offset noted; replacement planned

Remarks: type K thermocouples drift in this service. A type N element is proposed.

AT-1224 Treated effluent pH probe (loop AIC-1224, U-900)

Work order	Date	Found	Done
WO-25-0201	2025-01-28	0.4 pH above the lab	Probe cleaned and calibrated with pH 4 and 7 buffers
WO-25-1309	2025-06-11	0.5 pH above the lab	Probe replaced
WO-26-0119	2026-01-20	0.3 pH above the lab,	Reference junction

Work order	Date	Found	Done
		slow response	aged, probe replaced

PT-1008 Header pressure transmitter (loop PIC-1008, U-900)

Work order	Date	Found	Done
WO-25-0845	2025-04-23	1.5 % of span drift against the test gauge	Zero trimmed
WO-25-2101	2025-11-03	1.2 % drift	Zero trimmed; calibration interval reduced to 6 months

LV-1004 Product separator level valve (loop LIC-1004, U-200)

Work order	Date	Found	Done
WO-25-1420	2025-06-30	Operators report OP at 100 % and level high	Valve opens fully at 100 % OP, travel and supply pressure normal. Valve capacity insufficient at current throughput. Engineering request ER-25-044 for a larger trim

AT-1205 Hydrogen purity analyzer (loop AIC-1205, U-200)

Work order	Date	Found	Done
WO-25-1755	2025-09-01	Noisy reading	Sample filter clogged, replaced; sample flow adjusted to 1.0 L/min
WO-26-0350	2026-03-25	Noise returns	Sample line heat tracing faulty, condensation in the line. Repair planned